



RIVER VALLEY HIGH SCHOOL

YEAR 6 PRELIMINARY EXAMINATION

CANDIDATE
NAME

CENTRE
NUMBER

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INDEX
NUMBER

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CLASS

H1 BIOLOGY

8875/01

Paper 1 Multiple Choice

27 September 2013

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, Centre number and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate **Answer Sheet**.

Read the instructions on the Answer Sheet carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done on the Question Paper.

Calculators may be used.

This Question Paper consists of **20** printed pages.

For each question, there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

1. A ribosome consists of a large and a small subunit, each subunit containing ribosomal RNA (rRNA) complexed with protein.

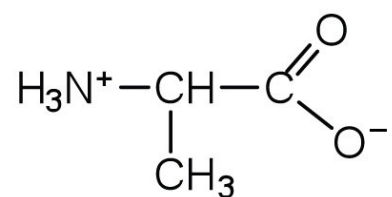
Which sequence of events concerning ribosomes is correct?

- A** Within the nucleolus rRNA and protein are synthesised and subunits formed. They become membrane bound as they are exported through the nuclear envelope to the cytoplasm and RER.
- B** rRNA and protein are synthesised in the Golgi body and are transported to the nucleolus for subunit formation.
- C** rRNA is synthesised in the SER and proteins are synthesised by the RER. Subunit formation occurs within the cytoplasm for free ribosomes and within RER for attached ribosomes.
- D** rRNA synthesised within the nucleus is complexed with protein that has been imported from cytoplasm. The subunits formed are exported to the cytoplasm via the nuclear pores.

2. Which type of molecular association is responsible for the α -helix configuration of polypeptide chains?

- A** Linkages between amino groups and carboxyl groups with the removal of water
- B** Bridges formed by H bonds established within the molecule between NH group of each amino acid residue and the CO group of another
- C** Ionic interactions between polypeptide chains
- D** Intra-chain bridges formed disulphide groups of the same chain.

3. The diagram below shows the structure of an amino acid.



Which of the following are true about this amino acid?

1. Non polar R group
2. Hydrophobic R group
3. Insoluble in water
4. Soluble in water

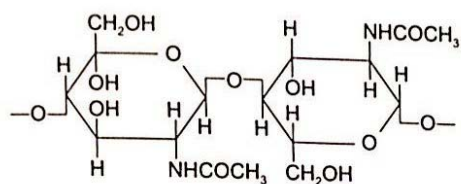
A 1 and 2 only

B 1, 2 and 3

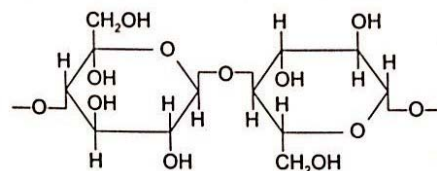
C 1, 2 and 4

D All of the above

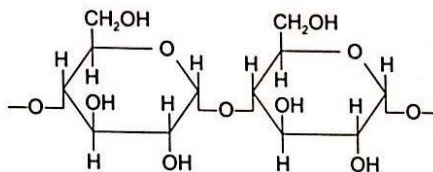
4. The diagrams show short sections of some common polysaccharides and modified polysaccharides.



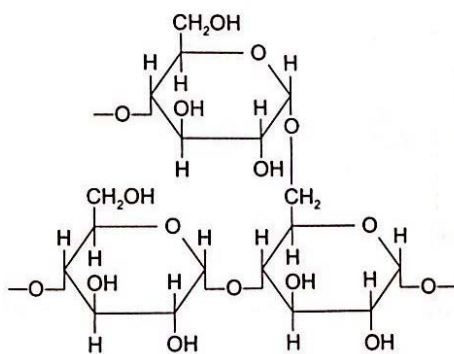
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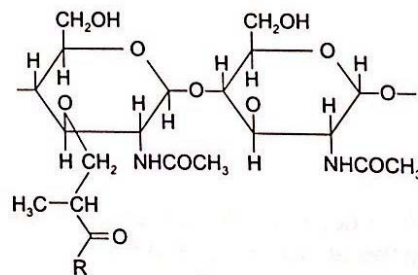
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3



4



5

The polysaccharides can be described as below.

- Polysaccharide F is composed of β -glucose monomers with 1,4 glycosidic bonds
- Polysaccharide G is composed of α -glucose monomers with 1,4 and 1,6 glycosidic bonds
- Polysaccharide H is composed of N-acetylglucosamine and N-acetylmuramic acid monomers with β -1,4 glycosidic bonds
- Polysaccharide J is composed of α -glucose monomers with 1,4 glycosidic bonds
- Polysaccharide K is composed of N-acetylglucosamine monomers with β -1,4 glycosidic bonds

Which shows the correct pairings of polysaccharide descriptions and diagrams?

	Polysaccharide F	Polysaccharide G	Polysaccharide H	Polysaccharide J	Polysaccharide K
A	2	4	5	3	1
B	2	5	4	1	3
C	3	4	1	2	5
D	3	5	4	1	2

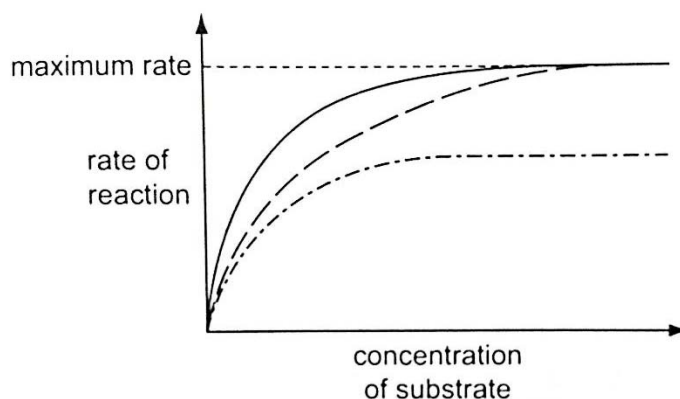
5. Catechol is oxidised to a benzoquinone, as shown in the equation, resulting in darkening of peeled fruits.



Catechol oxidase is an enzyme which is inhibited by parahydroxybenzoic acid (PHBA) which is structurally similar to catechol.

It is also inhibited by phenylthiourea (PTU) which binds to a copper atom in the enzyme. The copper atom is essential for the oxidative activity.

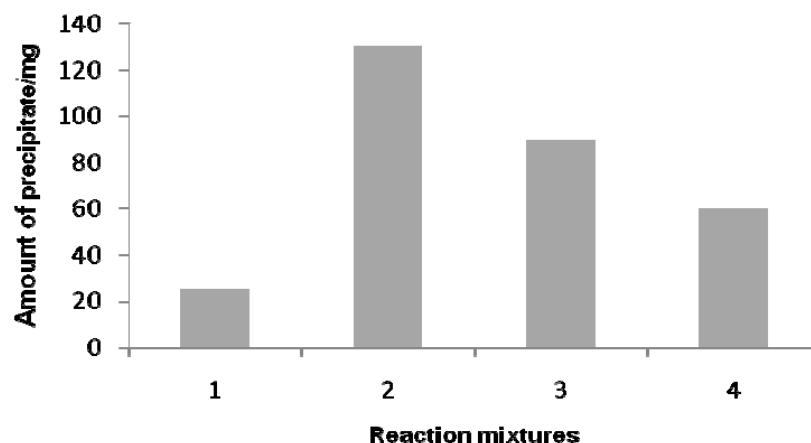
The graph shows the rate of the reaction with and without inhibitors.



Which is the correct key to the curves?

	with PHBA	with PTU	uninhibited
A		-----	————
B	————		-----
C	-----		————
D	-----	————	

6. Benedict's Test is often used to determine the presence and relative quantity of reducing sugars in samples. In the presence of reducing sugars, red copper oxide precipitate is obtained. The higher the concentration of reducing sugars, the greater the amount of red precipitate obtained. The graph below shows the amount of precipitate measured in reaction mixtures under different experimental conditions.



- 1 2% enzyme X + 10% starch + 28°C
- 2 2% enzyme X + 10% glycogen + 2% HCl + 100°C
- 3 2% enzyme X + 10% amylose + 2% HCl + 100°C
- 4 2% enzyme X + 10% starch + 60°C

Based on the data shown above, which conclusion is valid?

- A Change in pH is essential to the enzyme activity.
- B Rate of reaction is not affected by the temperature.
- C The degree of branching in substrate molecule is inversely proportional to the rate of reaction.
- D Many cysteine residues are present in the enzyme.

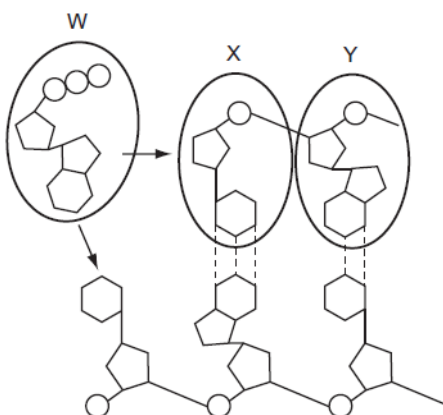
7. Messelson and Stahl found that in dividing cells, DNA is copied by semi-conservative replication. At the time of their discovery it was thought that DNA might be copied in one of three ways: semi-conservative, conservative and dispersive.

The bacteria were grown in culture containing heavy nitrogen, ^{15}N , until all the DNA was labelled. These bacteria were then transferred to a medium containing only light nitrogen, ^{14}N and allowed to replicate for two generations.

Which row shows the correct proportion of each type of DNA if DNA replication was conservative?

	All ^{14}N	Half ^{14}N , half ^{15}N	All ^{15}N
A	50%	50%	0%
B	0%	100%	0%
C	50%	0%	50%
D	75%	0%	25%

8. The diagram shows the synthesis of a polynucleotide. Molecule W is a nucleotide triphosphate.



Which statements are correct?

1. The base in W could be the purine, adenine
2. The base in Y is the purine, guanine
3. The base in X is the pyrimidine, cytosine
4. The base in X could be the pyrimidine, uracil

- A** 1 and 3 only
B 2 and 3 only
C 2 and 4 only
D 1, 2, 3 and 4

9. Which statements about the genetic code are correct?
1. There is only one codon for the amino acid methionine.
 2. Codons act as 'stop' and 'start' signals during transcription and translation.
 3. Prokaryotes generally use the same genetic code as eukaryotes.
 4. Stop codons are 'UAA, UGG, UGA'.
 5. mRNA codons have the same nucleotide sequence as DNA triplet codes.
- A 1 and 2 only
- B 1 and 3 only
- C 1, 3 and 4
- D 2, 4 and 5
10. What do chromosomal aberrations and gene mutations have in common and how are they different?

	Similarity	Difference
A	Both may involve addition of nucleotides	Gene mutations always produce dominant alleles but not chromosomal aberrations.
B	Both may not result in disorders	Gene mutations do not involve inversions but inversions of segments of chromosomes do occur.
C	Both affect nucleotide sequences in DNA molecule.	Gene mutations occur within a chromosome but chromosomal aberrations may occur across chromosomes.
D	Both may not result in a difference in protein expression.	Chromosomal aberrations are more harmful than gene mutations.

11. The petals of the morning glory flower are purple when the flower is in bud but become blue as the flower opens. The colour change is caused by an increase in pH of the vacuoles of the petal cells.

A mutation in a gene coding for a type of hydrogen ion (H^+) pump in the vacuole membranes of the petal cells results in flowers that are purple when open.

The difference in DNA sequence of the gene coding for the ion pump in plants with blue or purple flowers is shown below.

plant with blue flowers	-T-T-A-A-T-C-C-T-G-A-G-A-T-T-T-
plant with purple flowers	-T-T-A-A-T-C-C-T-G-C-T-G-A-G-A-T-T-T-

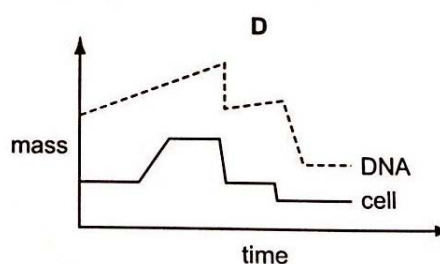
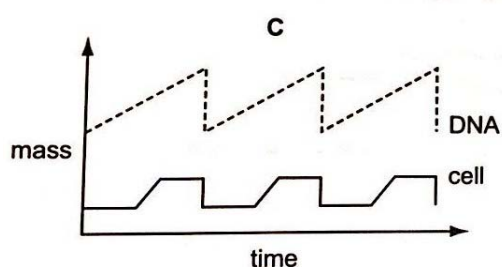
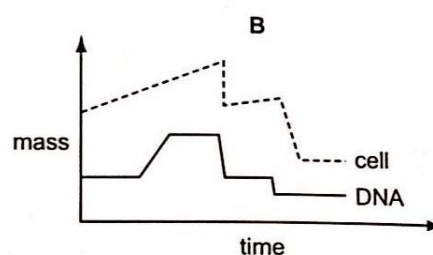
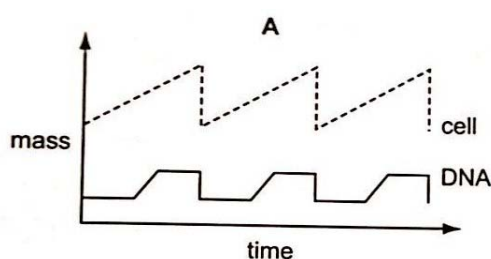
Which statements explain the purple colour of the flowers of the mutant plant?

1. A frameshift mutation alters the sequence of amino acids in the ion pump.
2. The primary structure and the shape of the ion pump are altered.
3. The activity of the ion pump is altered by changes in its shape.
4. The mutant ion pump cannot pump H^+ ions into the vacuoles of the petal cells

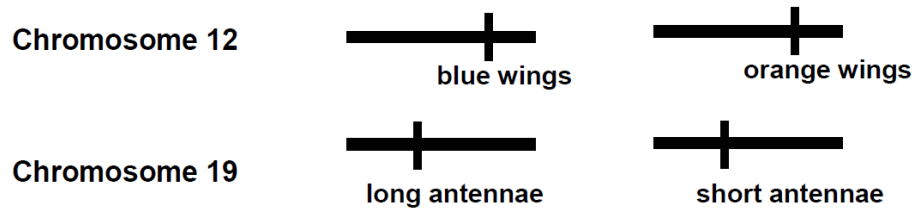
- A 1 and 2
B 1 and 4
C 2 and 3
D 3 and 4

12. Cell division is the means of almost all growth and reproduction.

Which graph correctly represents a form of cell division that maintains genetic stability at expense of variation?



13. A certain (hypothetical) organism is diploid, has either blue or orange wings as the consequence of one of its genes, and has either long or short antennae as a result of a second gene, as shown below.



If an organism is heterozygous for both genes, what will be the possible chromosomes and alleles found in one of her eggs?

- A Two chromosomes 12 with blue and orange wings alleles.
 - B Chromosome 12 with an orange wings allele and chromosome 19 with long antennae allele.
 - C Chromosome 19 with both orange wings allele and short antennae allele due to crossing over.
 - D Chromosome 12 with blue and orange wings alleles and chromosome 19 with long and short antennae alleles.
14. Which of the following is a characteristic of cancer cells?
- A Cellular differentiation
 - B Loss of contact inhibition
 - C Decreased oxygen uptake
 - D Inability to dislodge into blood streams

15. The table shows the results of a series of crosses in a species of small mammal.

coat colour phenotype		
male parent	female parent	offspring
dark grey	light grey	dark grey, light grey, albino
light grey	albino	light grey, white with black patches
dark grey	white with black patches	dark grey, light grey
light grey	dark grey	dark grey, light grey, white with black patches

What explains the inheritance of the range of phenotypes shown by these crosses?

- A one gene with a pair of co-dominant alleles
- B one gene with multiple alleles
- C sex linkage of the allele for grey coat colour
- D two genes, each with a dominant and recessive allele
16. In mice, the gene for 'dappled' coat (**D**) and its recessive allele for 'plain' coat (**d**), are located on the X chromosome. The gene for 'straight' whiskers (**W**) and its recessive allele for 'bent' whiskers (**w**) are autosomal.

A male mouse with plain coat and bent whiskers was mated on several occasions to the same female and the large number of offspring consisted of males and females in equal numbers in all combinations of phenotypes, as shown in the table.

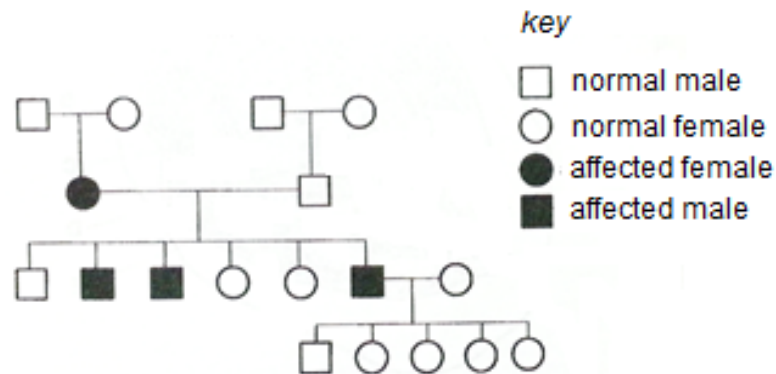
Offspring
Dappled, straight whiskers
Dappled, bent whiskers
Plain, straight whiskers
Plain, bent whiskers

If this female mouse is mated to a male with dappled coat and bent whiskers.

What proportion of the offspring would have at least one dominant allele at each gene locus?

- A $\frac{1}{8}$
- B $\frac{3}{8}$
- C $\frac{1}{4}$
- D $\frac{3}{4}$

17. The family tree shows the inheritance of a human defect.

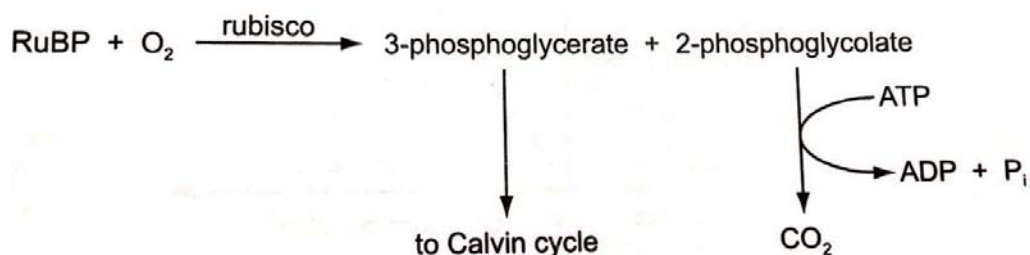


What is the most likely genetic basis for the occurrence of this defect?

- A Autosomal recessive
 - B Autosomal dominant
 - C Sex-linked recessive
 - D Sex-linked dominant
18. Which statement concerning chrysanthemum plants, of the genus *Dendranthema*, is a valid example of how the environment may affect the phenotype?
- A Anthocyanins and anthoxanthins are vacuolar pigments, whereas xanthophylls and carotenes are pigments found in membrane-bound organelles known as plastids. These, together with molecules known as co-pigments, are responsible for the variation observed in petal colour in *Dendranthema*.
 - B Identical genetic crosses performed between varieties of *Dendranthema* result in a greater proportion of offspring plants with plastids exhibiting a yellow colour when grown in a field and a greater proportion of offspring plants with colourless plastids when grown in a glass house.
 - C The seeds of a cross between *Dendranthema weyrichii* and *Dendranthema grandiflora* produce plants that are far more frost-tolerant and exhibit an extended flowering season compared with both parent plants.
 - D The seeds of a cross between *Dendranthema weyrichii* (height varying between 12.5 – 15.0 cm) and *Dendranthema grandiflora* (height varying between 8.0 – 25.0 cm) produce plants, when grown in natural day length, of a height varying between 55.0 – 71.0 cm.

19. Rubisco is the carbon dioxide-fixing enzyme. One rubisco molecule has eight active sites where carbon dioxide fixation occurs, with each active site catalysing only three reduction reactions per second.

The enzyme also catalyses, at the same active sites, the addition of oxygen to rubisco biphosphate (RuBP). This reaction is favoured when oxygen concentrations in the leaf are high and carbon dioxide concentrations are low.



Which of the facts is paired with a correct explanation?

	Fact	Explanation
A	On very hot, dry days stomata close to prevent water loss.	This reduces the availability of oxygen, increasing the production of 3-phosphoglycerate.
B	Plants synthesise large volumes of rubisco.	This may be an adaptive response to compensate for low concentrations of oxygen.
C	Processing 2-phosphoglycolate will eventually release carbon dioxide.	This will increase the rate of reduction and increase the rate of RuBP regeneration, increasing the rate of photosynthesis.
D	Rubisco is an inefficient photosynthetic enzyme.	This is because the rate of carbon dioxide reduction can be decreased by the competitive binding of oxygen molecules to the active site.

20. The table below shows reactions occurring in a plant cell, and their respective locations.

	Reaction	Location in a cell
1	ribulose biphosphate + CO ₂ → glycerate-3-phosphate	Stroma
2	glucose + ATP → glucose-6-phosphate + ADP	Matrix
3	oxygen + 4H ⁺ + 4e ⁻ → 2 H ₂ O	Stroma
4	oxaloacetate + acetyl-CoA → citrate	Matrix

Which of the following is / are **incorrectly** matched?

- A 2 only
- B 4 only
- C 2 and 3 only
- D 1, 2 and 3
21. Which of the following statement(s) about membranes is correct?
1. All intracellular membranes in a eukaryotic cell have the same type of lipids and proteins.
 2. The outer and inner membranes of mitochondria have the same type of transport proteins.
 3. Carbohydrates form part of glycoproteins or glycolipids in the membranes.
 4. All plant cell membranes have cholesterol.
- A 1 only
- B 3 only
- C 3 and 4 only
- D 1, 3 and 4

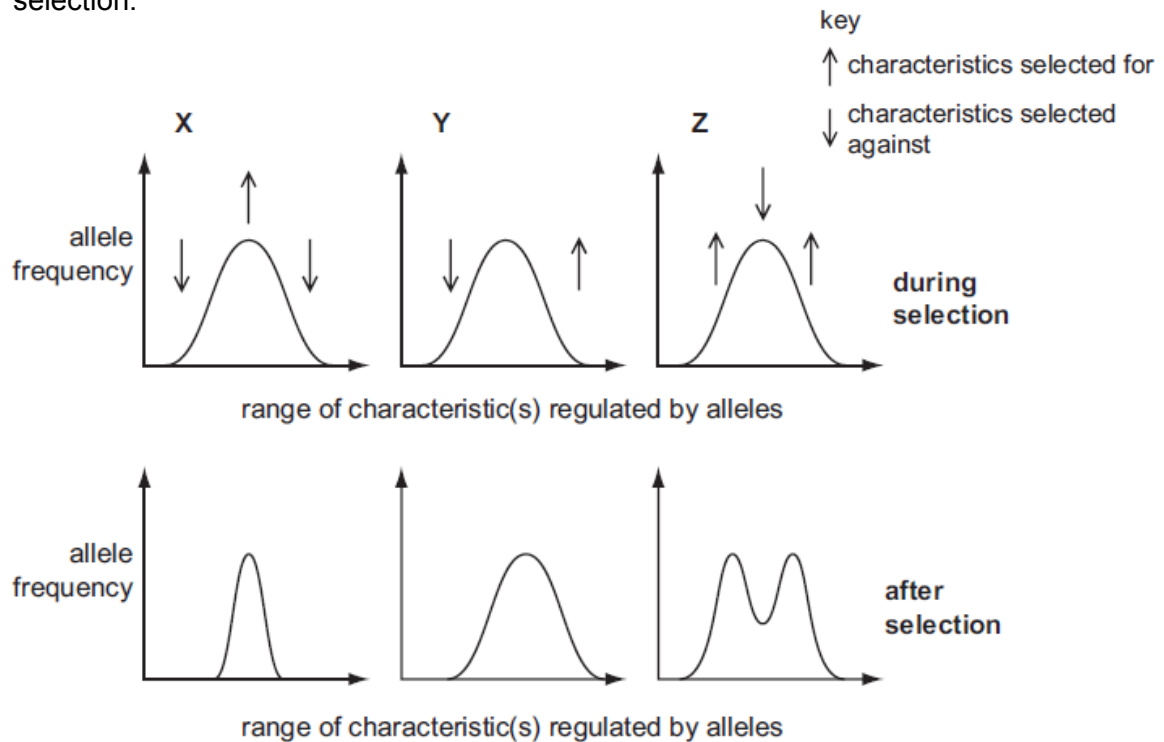
22. In an experiment on osmosis in plant cells, the solute potential (Ψ_s), pressure potential (Ψ_p) and water potential (Ψ_w) of the cell were recorded in the table below.

Condition of plant cell	Potential / kPa		
	Ψ_s	Ψ_p	Ψ_w
Plasmolysed	-1000	P	Q
Fully turgid	-600	R	S

Which of the following correctly represents the value of **P**, **Q**, **R** and **S**?

	P	Q	R	S
A	+1000	-2000	+600	0
B	-1000	-2000	0	-600
C	0	-1000	-600	-1200
D	0	-1000	+600	0

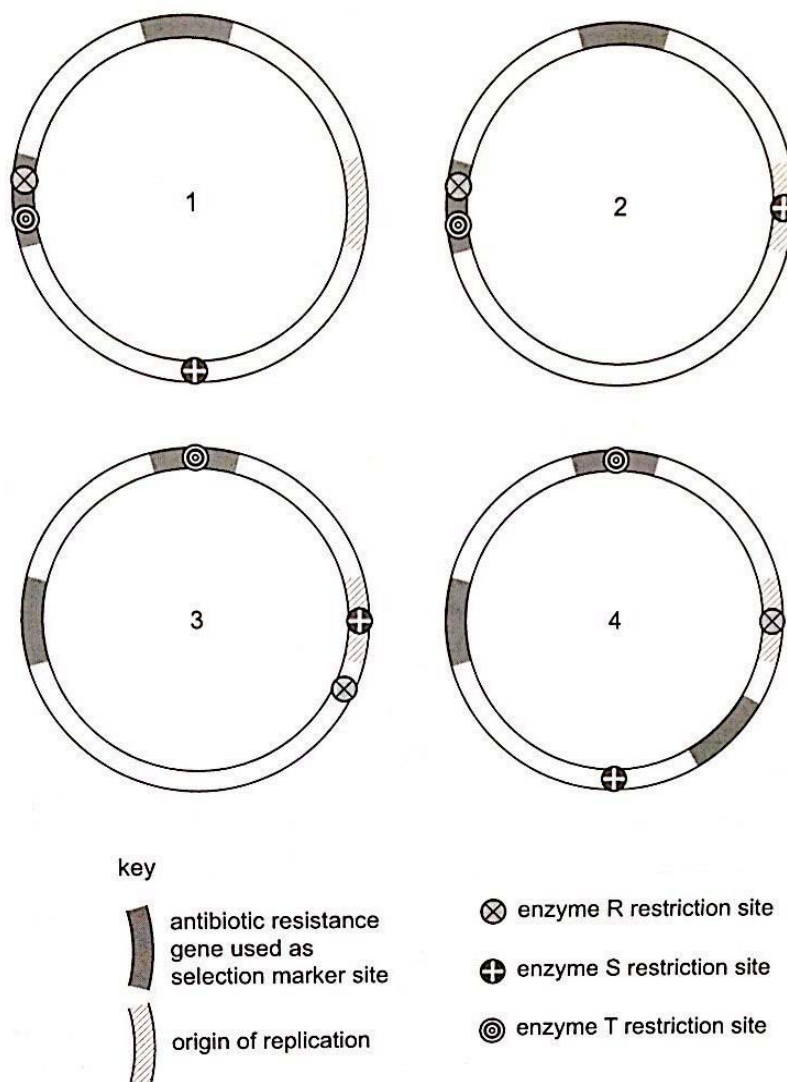
23. The graphs represent the frequency of alleles in species, X, Y and Z during the after selection.



In which species does evolution take place?

- A X only
 - B Y only
 - C Y and Z
 - D None of X, Y nor Z
24. *Lucilia cuprina*, the sheep blowfly, lays its eggs in wounds and the wet fleece of sheep. The larvae hatch and burrow into the sheep's skin, causing distress, reduced wool production and sometimes death. Chemicals were used in the past to control the *L. cuprina* but these became less effective as sheep blowfly developed a resistance to the chemicals. The cause of the increased resistance to the chemicals was most likely due to
- A susceptible sheep dying and leaving behind the more resistant sheep which selects for the more resistant blowfly.
 - B the insecticide producing a change in a gene which enhanced the survival of the blowfly.
 - C a chance mutation in a blowfly gene conferring a survival advantage in the chemical environment.
 - D the insecticide producing a change in phenotype which enhanced reproduction of the blowfly.

25. The diagram shows four plasmids.

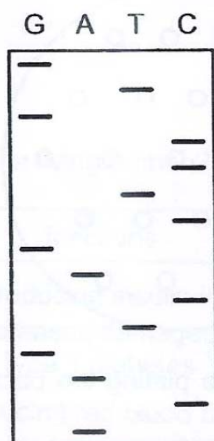


Which plasmids would be suitable to use for genetic engineering using restriction enzyme S?

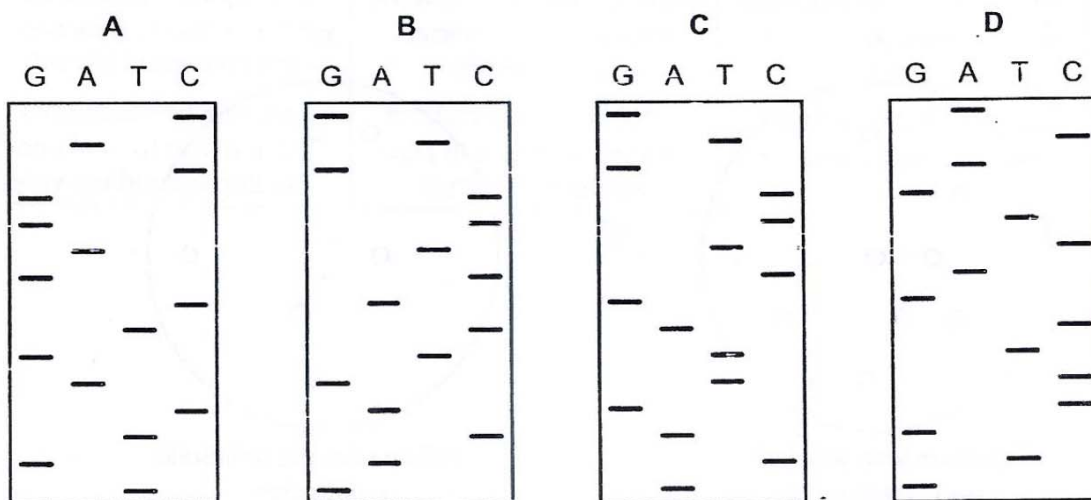
- A 1 and 4 only
- B 2 and 3 only
- C 1, 2 and 4
- D 1, 3 and 4

26. In people with a mutant allele, a protein contains just one different amino acid in its primary structure. To identify the presence of this mutant allele, DNA nucleotide sequences were compared using electrophoresis.

The electrophoresis results from the DNA of a normal allele of a gene are shown below.



Which diagram represents the DNA sequence for the mutant allele of this gene?



27. Which of the following is **not** necessary for Polymerase Chain Reaction to occur?

1. dATP
2. RNA primers
3. DNA fragment
4. Ribonucleotides

- A 1 only
- B 2 and 4
- C 1, 3 and 4
- D All of the above

28. In parallel with the Human Genome Project, the DNA of a set of model organisms (such as bacteria, roundworms, fruitflies, mice and chimpanzees) are being studied.

Which technique might be most useful in using the information from model organisms to provide critical clues about the structure and function of human genes?

- A Sequencing
- B DNA fingerprinting
- C Comparing molecular homology
- D Finding the regions of repetitive sequences

29. A key feature of most multicellular organisms is the ability to differentiate and produce specialised cells.

Which row best describes the ability of zygotic cells to differentiate?

	totipotent	pluripotent	multipotent
A	✓	✓	✓
B	✓	✗	✓
C	✓	✗	✗
D	✗	✓	✓

Key

✓ = ability

✗ = no ability

- 30.** Marker genes are often inserted into genetically engineered crop plant cells, along with desired genes. Bacterial antibiotic resistance genes are sometimes used as marker genes. These may include short DNA repeats to make them unstable so that they are quite quickly eliminated by the genetically engineered crop plant cells.

Which is not a reason why elimination of such marker genes is favoured?

- A** It is theoretically possible for the antibiotic resistance marker gene in human food to pass to bacteria in the human gut.
- B** It is difficult to carry out repeated transformations using the same antibiotic.
- C** The antibiotics may affect the growth and differentiation of the fields of crop plants.
- D** There are a few such antibiotic resistance marker genes available.